

DIABETES RISK PREDICTION & RISK SCORING

GROUP 6:

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PROBLEM DEFINITION

WHAT PROBLEM ARE WE SOLVING?

- Diabetes is one of the fastest growing chronic diseases worldwide. Early detection can significantly reduce complications and improve the quality of life of patients.
- Many individuals remain undiagnosed until serious health issues arise. Traditional diagnosis often requires medical tests that may not always be readily available.



Our project aims to develop a Machine Learning model that predicts whether a person is diabetic, pre-diabetic, or non-diabetic using health, lifestyle and demographic indicators.

OBJECTIVES:

- Predict diabetes risk
- Compare multiple ML algorithms
- Using Deep Learning algorithms
- Generate a personalized risk score
- Provide explainable predictions using SHAP

WHY THIS PROBLEM?

Diabetes affects millions of people globally.

Early prediction helps:

- Reduce healthcare costs
- Enable early lifestyle intervention
- Assist doctors in decision making
- Improve patient awareness

APPLICATIONS:

- Hospitals
- Healthcare apps
- Rural healthcare
- Preventive medicine



SIGNIFICANCE

- Early detection helps prevent severe diabetes-related complications.
- Supports timely medical intervention and lifestyle modifications
- Assists healthcare professionals in making data driven decisions.

MOTIVATION

- Diabetes is a rapidly growing global health concern
- Existing studies often overlook class imbalance and fair evaluation.
- To develop an accurate, explainable, and reliable diabetes prediction system.



EXPECTED OUTCOMES



The project will deliver the following outputs to support accurate prediction, risk assessment and clinical interpretability.



1. Diabetes Prediction

Classify individuals as Non-Diabetic, Pre-Diabetic or Diabetic.



2. Risk Score (0–100)

Generate a personalized diabetes risk score based on prediction probability.



3. ROC Curves

Compare the performance of all classifiers using ROC curves and AUC.



4. Confusion Matrix

Evaluate model performance using confusion matrix for each classifier.



5. Feature Importance

Identify and visualize the most influential features affecting diabetes risk.



6. SHAP Summary Plot

Provide model explainability using SHAP values for clinical interpretability.



7. BMI vs Age Analysis

Visualize the relationship between BMI and Age across different classes.



8. Risk Score Distribution

Show the distribution of risk scores across the population.



These outcomes will help in accurate prediction, better decision making and early intervention for improved patient care.



DATASET

DATASET: CDC Diabetes Health Indicators (BRFSS2015), sourced from the CDC's Behavioral Risk Factor Surveillance System survey.



251,680 survey responses, 21 health-related feature variables plus the target class.



Mainly categorical/binary features (e.g., HighBP, Smoker, PhysActivity), with a small number of continuous variables (BMI, MentHlth, PhysHlth).



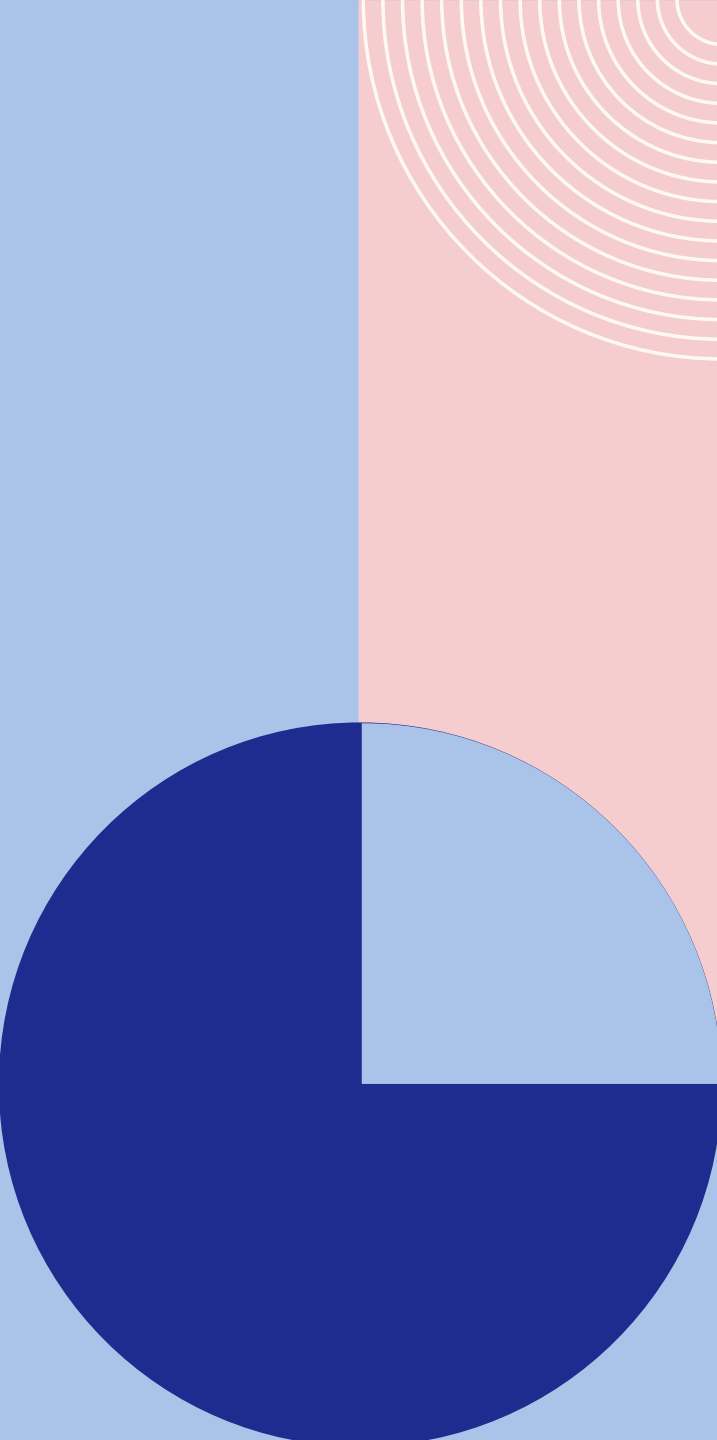
Three-class target: 0 – Healthy, 1 – Prediabetic, 2 – Diabetic.



~24,000 duplicate survey responses removed, leaving ~229,781 usable rows.

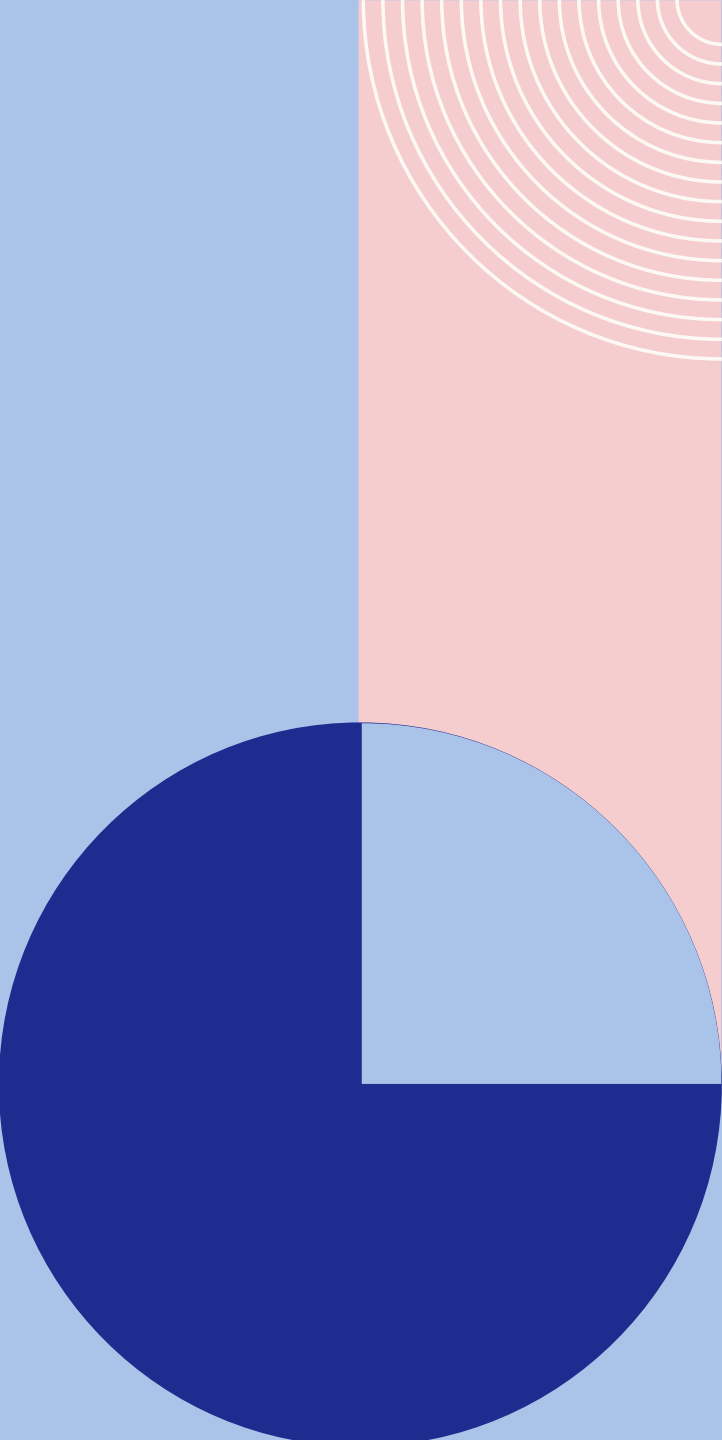


Severe class imbalance: roughly 86% Healthy, 14% Diabetic, and only ~1.8% Prediabetic – this imbalance is the central challenge of the entire project.



DATASET FEATURES

FEATURE	DESCRIPTION
HighBP	High Blood Pressure
HighChol	High Cholesterol
BMI	Body Mass Index
Smoker	Smoking Habit
Stroke	Stroke History
HeartDisease/Attack	Heart Disease
PhysActivity	Physical Activity
Fruits	Fruit Consumption
Veggies	Vegetable Consumption
HvyAlcoholConsump	Heavy Alcohol Consumption



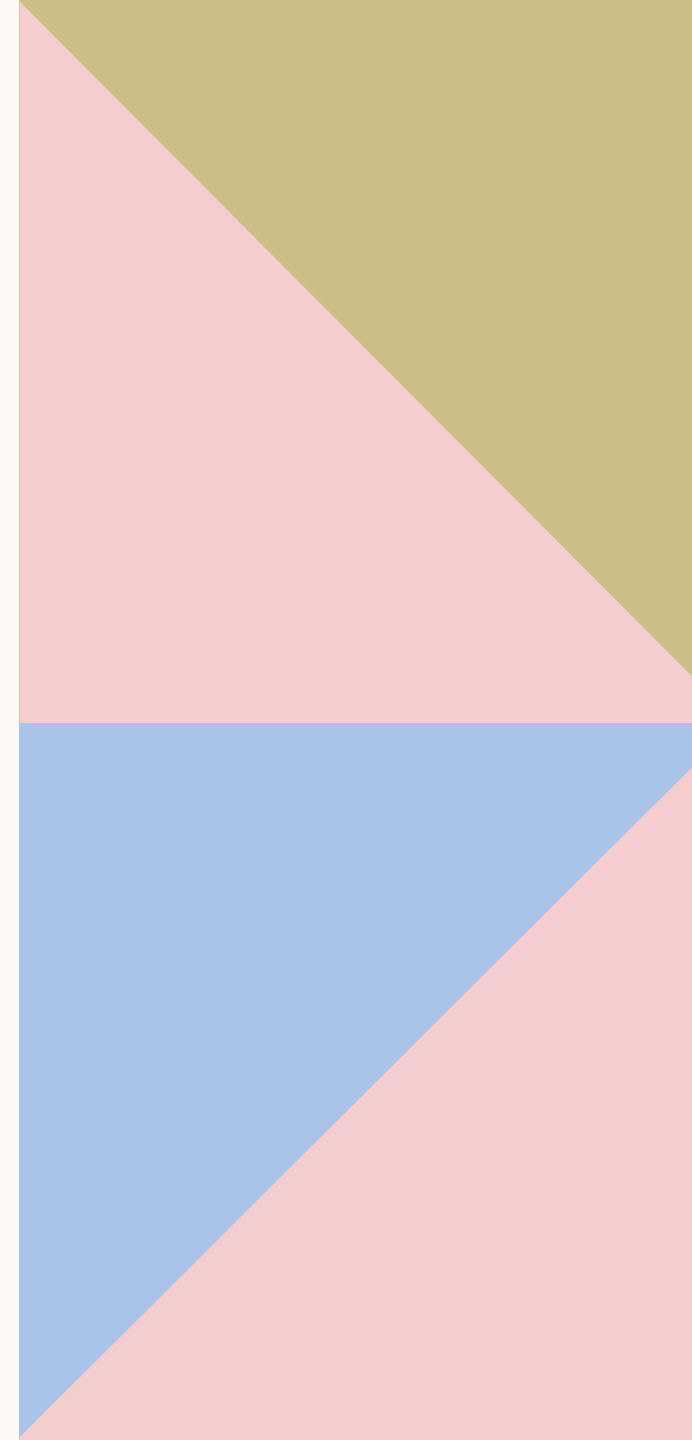
DATASET FEATURES

FEATURE	DESCRIPTION
AnyHealthCare	Healthcare Access
GenHlth	General Health
MentHlth	Mental Health
PhysHlth	Physical Health
DiffWalk	Difficulty Walking
Sex	Gender
Age	Age Category
Education	Education Level
Income	Income Level

Most features are binary (0/1). Some are ordinal and categorical.

DATASET CHALLENGES

- Class imbalance
- Missing lifestyle information
- Survey-based data may contain bias
- Many categorical variables
- Feature correlation
- Need for preprocessing and encoding



LITERATURE REVIEW – PAPER 1



PAPER 1



Available online at www.sciencedirect.com

Procedia Computer Science 132 (2018) 1578–1585

Procedia
Computer Science

International Conference on Computational Intelligence and Data Science (ICCIDIS 2018)

Prediction of Diabetes using Classification Algorithms

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Abstract

Diabetes is a chronic disease that occurs either when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces. Diabetes is one of the most common diseases, now a days. Data mining techniques are used to predict diabetes. In this paper, we have applied different classification algorithms on Pima Indians Diabetes dataset to predict diabetes. We have used algorithms like Decision Tree, Naïve Bayes and Support Vector Machine and compared its accuracy.

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Peer-review under responsibility of the scientific committee of the International Conference on Computational Intelligence and Data Science (ICCIDIS 2018).



OBJECTIVE

Develop a machine learning model for early diabetes prediction using patient health data.



ALGORITHMS USED

- Decision Tree
- Support Vector Machine (SVM)
- Naïve Bayes



KEY FINDINGS

- Naïve Bayes achieved the highest accuracy **(76.3%)** on the Pima Indians Diabetes Dataset.
- Machine learning can assist in early diagnosis and reduce the need for extensive medical testing.



Paper URL: <https://www.sciencedirect.com/science/article/pii/S1877050918308548>



DOI: 10.1016/j.procs.2018.05.122

LITERATURE REVIEW – PAPER 2

PAPER 2



Available online at www.sciencedirect.com

Expert Systems with Applications 78 (2017) 305–334



Machine Learning and Data Mining Methods in Diabetes Research

Konstantinos Kavakiotis, Ioannis Koumakis, Ioannis Maglogiannis,
Dimitrios Tzovaras, Constantinos Vassilakis

Abstract

Diabetes is a global health issue that requires early detection and effective management. This paper presents a comprehensive review of Machine Learning and Data Mining techniques applied in diabetes diagnosis and management. The study discusses various algorithms, datasets, evaluation metrics, and key findings from recent research. Machine Learning models can support early detection, personalized treatment, and better decision-making in healthcare.



OBJECTIVE

To review the existing Machine Learning and Data Mining approaches used in diabetes diagnosis and management, highlighting their effectiveness and limitations.



KEY FINDINGS

- Machine Learning techniques are effective for **early diabetes detection** and risk assessment.
- Data **preprocessing** and **feature selection** significantly improve model performance.
- **Ensemble learning methods** generally outperform traditional algorithms.
- ML models can aid in **personalized treatment** and better clinical decision-making.



Takeaway: Machine Learning and Data Mining techniques show great potential in diabetes diagnosis and management. Proper data handling and algorithm selection are key to achieving better accuracy and reliability.

LITERATURE REVIEW – PAPER 3



BASE PAPER



ELSEVIER

Engineering Reports
Volume 7, March 2025, e12803



Explainable Machine Learning for Efficient Diabetes Prediction Using Hyperparameter Tuning, SHAP Analysis, Partial Dependency, and LIME

Md. Najmul Islam  , Md. Anik Islam, Foysal Ahmed, Mohammad Monzurul Ahsan

Abstract

Diabetes is a major global health concern requiring accurate and interpretable prediction models. This study proposes an explainable machine learning framework that integrates hyperparameter tuning with SHAP, LIME, and Partial Dependence Plots to enhance prediction performance and interpretability. Using the CDC BRFSS2015 dataset (253,680 records, 22 features), the Extra Trees Classifier achieved the highest accuracy of 97.5% after optimal tuning. Explainable AI techniques provided clear insights into individual predictions, enabling greater transparency and trust in the model.



URL: <https://doi.org/10.1002/eng2.12803>

Wiley Online Library: <https://onlinelibrary.wiley.com/doi/10.1002>



OBJECTIVE

To develop an explainable diabetes prediction model by combining hyperparameter tuning, ensemble machine learning algorithms, and Explainable AI techniques (SHAP, LIME, and Partial Dependence Plots) for accurate and transparent predictions.



TECHNIQUES USED

- Hyperparameter Tuning
- Extra Trees Classifier
- Random Forest
- SHAP Analysis
- LIME
- Partial Dependence Plots (PDP)



KEY FINDINGS

- **Extra Trees** achieved the highest prediction accuracy (**97.5%**) after hyperparameter tuning.
- **SHAP** and **LIME** successfully explained individual model predictions, improving transparency.
- Explainable AI techniques increased **model interpretability** while maintaining high predictive performance.
- The proposed framework demonstrated **reliable diabetes risk** prediction using the CDC BRFSS2015 dataset.



Takeaway: Combining hyperparameter tuning with Explainable AI (SHAP, LIME, and Partial Dependence Plots) significantly improves both the accuracy and trustworthiness of diabetes prediction systems.

BASE PAPER SELECTION



The published study our project directly benchmarks against and extends



Explainable Machine Learning for Efficient Diabetes Prediction Using Hyperparameter Tuning, SHAP Analysis, Partial Dependency, and LIME

Md. Najmul Islam et al. • 2025 • Engineering Reports (Wiley), e13080

Dataset: CDC BRFSS2015 (253,680 records, 21 features) | **Best result:** Extra Trees, up to 97.5% accuracy after oversampling + tuning, with SHAP/LIME explainability

WHY WE CHOSE THIS AS OUR BASE PAPER

1



Same Dataset

Uses the identical CDC BRFSS2015 dataset (253,680 records, 21 features) as our project — directly comparable results.

2



Same Objectives

Compares multiple ML models, applies hyperparameter tuning, and uses SHAP + LIME — matching our core objectives exactly.

3



A Clear Gap

Its oversampling step isn't confirmed to be strictly post-split, and it omits deep-learning tabular models — a gap our project addresses.

4



Room to Extend

No deployment-ready tool is offered; our live Streamlit dashboard with SHAP-based risk factors builds directly on its findings.

[CLICK HERE TO VIEW THE SPREADSHEET](#)

EXISTING WORK – KEY INSIGHTS FOR OUR PROJECT



*Extracted insights from existing studies on CDC BRFSS2015 dataset
that shape and differentiate our project.*



1. STRENGTHS (FROM LITERATURE)

- ✓ Strong performance metrics and high reported **accuracy** across many models.
- ✓ Wide **exploration** of traditional ML and ensemble models.
- ✓ Ensemble and boosting methods generally show **robust** results.
- ✓ Deep learning / transformer models **rarely explored** – a gap our project addresses.



2. LIMITATIONS (FROM LITERATURE)

- ✗ Many studies apply oversampling techniques (Random Oversampling, SMOTE) before train/test split → causes **data leakage**.
- ✗ This leads to **inflated performance metrics** that do not reflect real-world generalization.
- ✗ Reported accuracies may therefore be **overestimated** and not reliable for real-world applications.
- ✗ Few studies provide a real-world, **deployment-ready, explainable tool**.



3. HIGH ACCURACY MODELS SUSPECTED OF OVERSAMPLING

01

KNN (with oversampling)

Reported accuracy:

98.38%

02

Extra Trees (with oversampling)

Reported accuracy:

97%

These results may be inflated due to oversampling before train/test split.

BASE PAPER vs OUR PROJECT



How our proposed pipeline builds on and improves the base paper

	BASE PAPER (Islam et al., 2025)	OUR PROJECT
 Class Imbalance	Random oversampling applied; split timing not clearly confirmed	SMOTE applied strictly AFTER the train/test split — no leakage
 Models Compared	KNN, Random Forest, Decision Tree, Extra Trees	Adds XGBoost, LightGBM, CatBoost, FT-Transformer, TabTransformer
 Model Ranking Metric	Primarily accuracy after tuning	Ranked by weighted F1-score , precision, recall, ROC-AUC
 Explainability	SHAP + LIME as static plots in the paper	SHAP surfaced live as patient-level “ Top Risk Factors ”
 Real-World Usability	No deployed tool provided	Live Streamlit dashboard : manual entry, PDF upload, model comparison



Novelty & impact: *leakage-free evaluation, broader model benchmarking (including tabular deep learning), fairer metric selection, and a deployment-ready explainable tool.*

LITERATURE REVIEW – COMPARATIVE ANALYSIS



- A comparative study of existing research papers related to diabetes prediction using machine learning.

 Author (Year)	 Type of Study	 Technique / Algorithm	 Major Result / Contribution
 Sisodia & Sisodia (2018)	Experimental Study	 Random Forest Algorithm	- Random Forest outperformed other classifiers with highest accuracy. - Demonstrated ML effectiveness for early diabetes prediction.
 Kavakiotis et al. (2017)	Review Paper	 Review of Multiple ML Methods	- Comprehensive review of ML & data mining methods in diabetes research. - Concluded ML techniques are highly useful for diagnosis and management.
 Islam et al. (2025)	Experimental Study	 Hyperparameter Tuning + SHAP + LIME + PDP + Extra Trees (Ensemble)	- Extra Trees achieved the highest accuracy (97.5%). - SHAP, LIME, and PDP improved interpretability and transparency. - Proposed explainable framework ensures both accuracy and trustworthiness.



Takeaway: While previous studies established the effectiveness of machine learning for diabetes prediction, our base paper (Islam et al., 2025) advances the field by integrating hyperparameter tuning and explainable AI techniques, achieving **high accuracy** with **interpretability**.

LITERATURE REVIEW



RESEARCH GAP / LIMITATIONS



1. Mostly Binary Classification

Many studies focus on only two classes (Diabetic / Non-Diabetic) instead of multi-class prediction (Non-Diabetic, Pre-Diabetic, Diabetic).



2. Limited Interpretability

Most models act as black boxes, making it difficult to understand why a particular prediction is made.



3. Lack of Personalized Risk Assessment

Very few studies provide a personalized diabetes risk score that can help in proactive healthcare decision-making.



4. Class Imbalance Often Ignored

Many works do not properly address class imbalance, which can lead to biased and unreliable predictions.



5. Focus on Accuracy, Not Usability

Most research prioritizes improving accuracy, but real-world usability, explainability, and deployment aspects are less explored.



Takeaway: Existing studies have made significant progress, but there is a clear need for a more interpretable, multi-class, and personalized diabetes prediction system with balanced evaluation and real-world applicability.

LITERATURE REVIEW



HOW OUR WORK ADDRESSES THESE GAPS



1. Multi-class Prediction

- Predicts three classes for better clinical relevance:



Non-Diabetic



Pre-Diabetic



Diabetic



2. Personalized Diabetes Risk Score

- Generates a personalized risk score to help users understand their risk level and take proactive measures.



Risk Score
72 / 100
High Risk



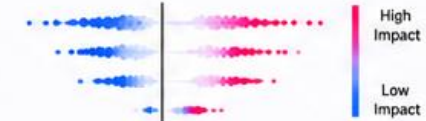
3. Comparison of Multiple Machine Learning Algorithms

- Compares the performance of multiple ML algorithms to identify the most accurate and reliable model.



4. Explainable AI using SHAP

- Uses SHAP (SHapley Additive exPlanations) to explain predictions and increase transparency and trust.



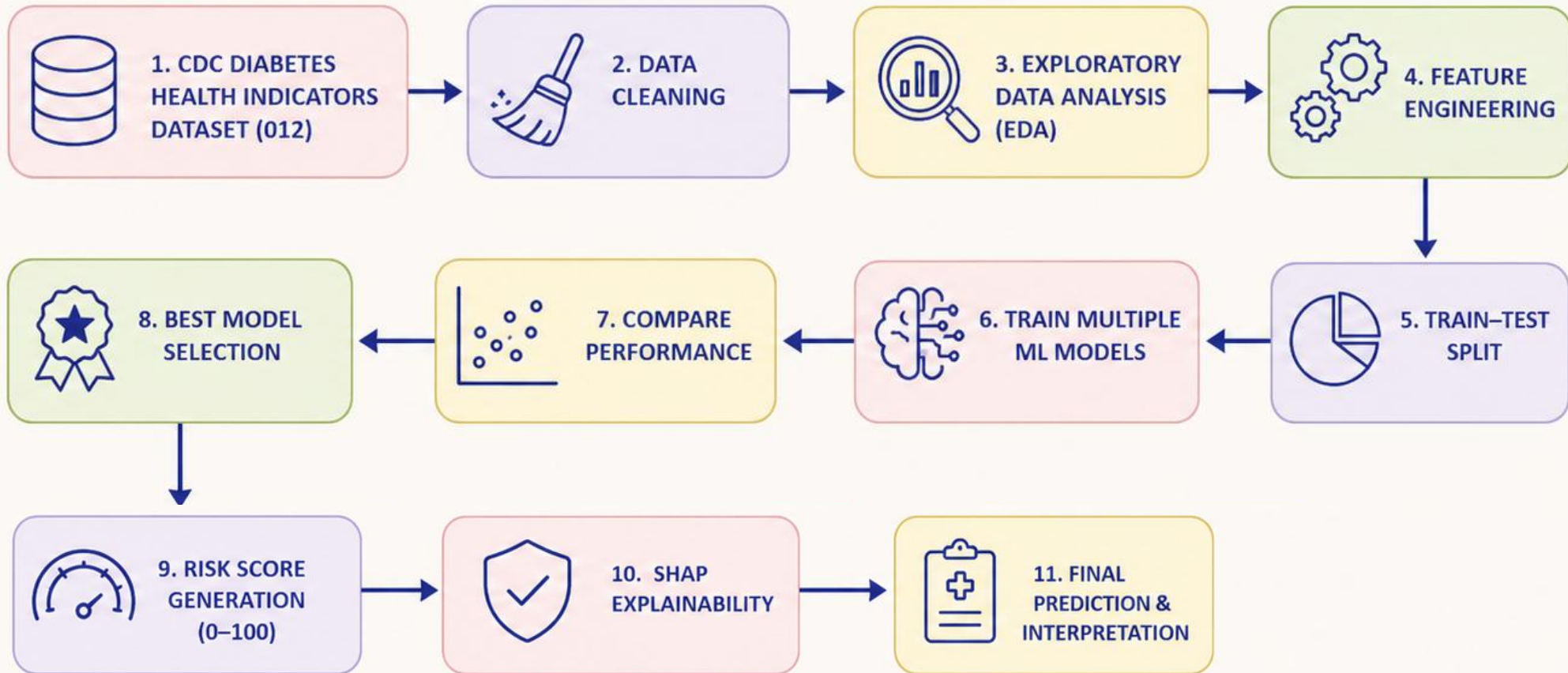
5. Better Preprocessing and Fair Evaluation

- Applies robust preprocessing, handles class imbalance, and uses fair evaluation metrics for reliable results.



Takeaway: Our approach aims to build an accurate, interpretable, and user-friendly diabetes prediction system that not only classifies but also empowers individuals with actionable insights.

PROPOSED METHODOLOGY



MACHINE LEARNING MODELS



MODELS USED



Logistic Regression



Random Forest



XGBoost



Light GBM



CatBoost



FT-Transformer



TabTransformer



EVALUATION METRICS



Accuracy



Precision



Recall



F1 Score



ROC-AUC



Confusion Matrix



Classification Report



Selected models will be compared to identify the best-performing classifier for diabetes risk prediction.

METHODOLOGIES & ADVANCED TECHNIQUES USED

Key techniques and advanced models applied in this project to ensure robust, fair and high-performing predictions.



1. SMOTE (Synthetic Minority Over-sampling Technique)

- ✓ Addresses severe class imbalance in the dataset.
- ✓ Generates synthetic samples for the minority class by interpolating between nearest neighbors.
- ✓ Applied ONLY on the training set after the train/test split to prevent data leakage.
- ✓ Helps models learn better decision boundaries and improves recall & F1-score for the minority class.



2. HYPERPARAMETER TUNING

- ✓ Performed using GridSearchCV / RandomizedSearchCV.
- ✓ Optimizes model performance by finding the best combination of hyperparameters.
- ✓ Improves generalization and reduces overfitting.
- ✓ Key parameters tuned for all models based on cross-validation scores.



3. FT-TRANSFORMER (For Tabular Data)

- ✓ A transformer-based model designed for tabular data.
- ✓ Captures complex feature interactions and long-range dependencies efficiently.
- ✓ Handles mixed data types (numerical + categorical) without extensive preprocessing.
- ✓ Provides a powerful deep learning alternative to traditional ML models.



4. TABTRANSFORMER

- ✓ Utilizes self-attention on categorical features while embedding them.
- ✓ Excels at learning relationships between categories and numeric features.
- ✓ Well-suited for structured/tabular datasets with high cardinality.
- ✓ Benchmarked against classical ML models to evaluate real-world performance.



Why these techniques? SMOTE ensures balanced learning, hyperparameter tuning maximizes performance, and advanced transformer models (FT-Transformer & TabTransformer) bring deep learning capabilities to tabular data — offering improved accuracy, generalization and interpretability.



Together, these methods build a **robust, fair and state-of-the-art prediction pipeline.**

PROPOSED APPROACH



A comprehensive 7-model comparison – **Logistic Regression**, **Random Forest**, **XGBoost**, **LightGBM**, **CatBoost**, **FT-Transformer**, and **TabTransformer** – evaluated with both **weighted and macro-averaged precision, recall, F1, and ROC-AUC** (not accuracy alone), combined with **SHAP-based explainability** and a **live Streamlit dashboard** supporting manual entry, PDF-report upload, and model comparison.

WHY THIS APPROACH, AND HOW IT IMPROVES ON EXISTING METHODS



01 METRIC SELECTION

- Models are ranked by **F1 Score (weighted)** rather than accuracy.
- Addresses the flaw of accuracy rewarding models that ignore the minority (diabetic) class.



02 IMBALANCE HANDLING

- SMOTE is applied to the training set only, strictly after the train/test split.
- Ensures honest evaluation without synthetic data leaking into the test set.



03 MODEL BREADTH

- Classical ML models (Logistic Regression, tree ensembles, gradient boosting) are benchmarked against modern deep learning models for tabular data (FT-Transformer, TabTransformer).
- A comparison largely absent in existing literature on this dataset.



04 EXPLAINABILITY

- SHAP-based feature attribution is generated for the top-performing model.
- Surfaced in the dashboard as patient-level "Top Risk Factors", not just a static plot in a report.



05 REAL-WORLD USABILITY

- Streamlit dashboard enables live predictions via manual entry or PDF health-report upload (with automatic field extraction).
- Includes full model comparison view – moving beyond a results table into an actual usable tool.



OVERALL IMPACT



More reliable and fair evaluation through better metrics and no data leakage.



Stronger, modern benchmark with deep learning models for tabular data.



Greater interpretability with SHAP-driven insights for real clinical relevance.



Deployment-ready dashboard for real-world use by healthcare stakeholders.

FUTURE WORK



The project can be further enhanced and extended to increase its accuracy, usability and real-world impact.



1. Web Application Development

Build an interactive web application using Flask for real-time predictions.



2. Real-time Prediction

Enable real-time risk prediction for instant results and feedback.



3. Integration with Wearable Devices

Integrate with health wearables to collect real-time data and monitor risk.



4. Advanced Models

Explore deep learning models to improve prediction accuracy.



5. Cloud Deployment

Deploy the model on cloud platforms for scalability and availability.



6. Mobile Application

Develop a mobile app for easy access and better user engagement.



7. Patient Personalization

Incorporate more patient specific data for personalized risk assessment.



8. Broader Impact

Use the system in rural healthcare and preventive healthcare initiatives.



These improvements will help build a more accurate, scalable and impactful system that can support early intervention and better healthcare decisions.

